

Prevalence of Congenital Malaria in an Urban and a Semirural Area in Lagos; a Two-centre Cross-sectional Study

Moyinolorun Oluwakayode Omidiji

moyinomidi.ji@yahoo.com

Lagos University Teaching Hospital

Afolabi Lesi

Lagos University Teaching Hospital

Christopher Esezobor

Lagos University Teaching Hospital

Iretiola Fajolu

Lagos University Teaching Hospital

Wellington Oyibo

Lagos University Teaching Hospital

Adetola Daramola

Lagos University Teaching Hospital

Article

Keywords: Congenital malaria, IPT-SP (Intermittent Preventive Treatment with sulfadoxine-Pyrimethamine, Parasitaemia, LLIN (long lasting Insecticide treated nets).

Posted Date: January 15th, 2025

DOI: <https://doi.org/10.21203/rs.3.rs-5759311/v1>

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Additional Declarations: No competing interests reported.

Version of Record: A version of this preprint was published at Scientific Reports on March 28th, 2025.
See the published version at <https://doi.org/10.1038/s41598-025-94800-w>.

Abstract

Background

Congenital malaria is a recognized cause of morbidity and mortality in newborns. Signs and symptoms of congenital malaria are non-specific and could be confused with Neonatal sepsis. There has been a recent decline in malaria burden worldwide attributed to a new strategy recommended by the WHO including the use of intermittent preventive treatment of malaria in pregnancy using Sulfadoxine-Pyrimethamine (IPT-SP) during pregnancy, long lasting insecticide treated nets (LLINs), malaria case management with Artemisinin-based combination therapy etc. This study sets out to determine the effect of this changes on the prevalence of congenital malaria in two centres in Lagos, Nigeria.

Methods

Using a cross-sectional observational descriptive design, a total of 291 mother and newborn pairs were enrolled from an urban area and a semi-rural area in Lagos between April and October 2014. About three-fifths of the total study population was derived from the urban centre. A predesigned questionnaire was used to extract basic physical and demographic information such as the use of IPT-SP during pregnancy. Malaria microscopy was carried out on the maternal blood samples, and, the corresponding newborns' heel prick and cord blood samples while the placenta tissues were examined for malaria pigments.

Results

Malaria parasitaemia, cord blood and congenital malaria were 0.34%, 0% and 0% respectively while that of placental malaria pigmentation was 18.9%. Placental malaria incidence was less in mothers who received IPT-SP in pregnancy ($p = 0.016$). Placental malaria incidence was higher in mothers ≤ 24 years ($p = 0.044$) and the less educated women had a higher prevalence of placental malaria ($p = 0.001$). The incidence of placental malaria was higher in the semi-rural area (92.7% v 7.3%, $p = < 0.0001$). Newborns of mothers with placental malaria had lower birth weight (2881.8 v 3100.7 g, $p = 0.020$) and smaller head circumference (34.3 v 35.1 cm, $p = 0.006$)

Conclusion-

This study demonstrated a significant decline in the prevalence of congenital malaria reflecting the recently reported decline in the burden of malaria in the general population in Africa. Use of IPT-SP during pregnancy, urban area residence and higher educational status appear to have been protective against malaria. A regular surveillance is however necessary considering the dynamics involved in malaria drug resistance.

BACKGROUND

Malaria remains one of the leading causes of childhood morbidity and mortality especially in the tropical countries of the world.¹ In 2022, there was an estimated 249 million malaria cases worldwide and

608,000 malaria deaths in 85 countries. The WHO African Region carries a disproportionately high share of the global malaria burden with the region being home to 94% of malaria cases (233 million) and 95% (580, 000) of malaria deaths. Children under 5 accounted for about 80% of all malaria deaths in the region. Four African countries accounted for just over half of all malaria deaths worldwide with Nigeria topping the list with 26.8%.²

Of the five species of malaria, *P. falciparum* is the most common in sub-Saharan Africa (SSA) and it is responsible for most of the severe complications of malaria in adults and children.^{3,4}

Malaria infection appears to be more prevalent in rural areas compared to the urban areas as demonstrated by existing studies.^{5,6} This has been attributed to factors such as land use and demographic factors like industrialization and urban agricultural activities which lead to the disruption of the mosquito larval habitat. In addition, easier access to health care facilities in the urban areas and a higher population density in the urban areas leading to a lower mosquito bite rate due to higher human-mosquito ratio are other factors thought to be responsible for the lower burden of malaria in urban areas compared to rural areas.

Malaria during pregnancy (MIP) is a public health concern because of the negative effects it can have, not only on the mother but also on the developing foetus.⁷ Pregnant women remain at high risk of MIP, with over 50% of pregnant women in high transmission areas having *P. falciparum* detected in peripheral blood at presentation to antenatal care.⁸ The physiological and hormonal changes during pregnancy result in general immunosuppression, making pregnant women more vulnerable to malaria. This immunosuppression, combined with the effect of the disease, creates a synergistic interaction that can lead to severe outcomes for both the mother and fetus.⁸ Pregnant women primarily exhibit type-2 cytokine responses (mediated by interleukins 10, 4, and 6), favoring humoral immunity over the type-1 cellular response (involving interferon-gamma, interleukins 2 and 12, and tumor necrosis factor). This shift helps protect the fetoplacental unit but leaves pregnant women more susceptible to infections like malaria, tuberculosis, and leishmaniasis, which rely on type-1 responses for immunity.⁹ Another theory links reduced lymphocyte activity to elevated levels of hormones such as cortisol, human chorionic gonadotropin (HCG), and progesterone, which, while preventing fetal rejection, increase vulnerability to infections.⁹

The placenta also plays a critical role in malaria susceptibility during pregnancy. As a new organ in first-time mothers, it provides a niche for malaria parasites, allowing placental-specific strains of *P. falciparum* to thrive. Over time, women develop immunity to placental malaria, which may explain why multigravidae are at a lower risk. McGregor et al,¹⁰ suggested that the placenta's unique environment favors malaria parasites. Parasitized red blood cells express surface molecules like chondroitin sulfate A (CSA) and hyaluronic acid, which help parasites adhere to placental structures, including trophoblastic villi and syncytial bridges, where they obstruct nutrient and oxygen flow to the fetus- a term referred to as

malaria sequestration. During active placental infection, villi size, shape, and vascularity decrease, further reducing gas exchange capacity.

Placental sequestration of malaria parasites can be responsible for the absence of malaria parasites peripheral blood, as supported by studies like that of Mateelli et al,¹¹ who found that many women had placental infection without detectable parasites in their blood. This sequestration prevents infected red cells from reaching the spleen, where they would typically be cleared.

The impact of malaria in pregnancy (MiP) varies by malaria transmission intensity in a region. Pregnant women in low-transmission areas, who lack immunity, face a two- to threefold higher risk of severe malaria complications compared to their non-pregnant peers. These complications include severe anaemia, hypoglycemia, acute pulmonary oedema, cerebral malaria, maternal death etc. The impact on the fetus may include premature delivery, stillbirth, low birth weight (LBW), congenital malaria or neonatal death. Conversely, women in high-transmission regions, where they have built partial immunity over time, often experience milder symptoms and may even be asymptomatic. In these areas, anemia and LBW are the primary issues in the infected mothers and neonates, respectively.^{11–13}

Congenital malaria (CM) is defined as the presence of malaria parasite in the newborn within the first seven days of life.^{14–16} It is a form of malaria a newborn acquires from the mother through the placenta either prenatally or during delivery.^{14,15,17} The common denominator among the various definitions of Congenital malaria by different researchers is that the malaria parasite is demonstrable in the peripheral blood of the new born within the first seven days of life.^{14–16,18} Runsewe- Abiodun *et al*,¹⁹ defined congenital malaria as symptoms attributable only to malaria with evidence of ring forms of malaria parasite in the erythrocyte of an infant within the first seven days of life. However, some studies have described both symptomatic and asymptomatic forms of congenital.^{14,20–21} It has also been defined as the detection of malaria parasite in the newborn within seven days of birth or later if there is no possibility of postpartum infection by either mosquito bite or blood transfusion.^{19,22} Congenital malaria can lead to complications such as miscarriage, premature birth, low birth weight (LBW), and neonatal anemia.^{23,24} Studies from Lagos, Minna and Abuja have reported prevalence of congenital malaria between 2.63% and 46.7% in the preceding two decades before this study,^{14,16,17,25} suggesting there had been a rise in the burden of congenital malaria compared to older studies.^{26–28} These reported surges coincided with the era of the peak of chloroquine resistance and the use of proguanil for malaria prevention during pregnancy.^{29,30}

However, through the concerted efforts of the Roll Back Malaria partnership and major donor foundations such as the Bill and Melinda Gates Foundation, there have been changes in the approach to the management malaria in terms of treatment protocol, chemo-prophylaxis and, consequently, a reduction in the burden of malaria.^{31,32} Intermittent preventive treatment with sulfadoxine-pyrimethamine (IPT-SP) for malaria chemo-prophylaxis during pregnancy replaced chloroquine (due to increased malaria resistance to chloroquine), proguanil and pyrimethamine.³⁰ Secondly, Artemisinin-based combinations

became the treatment of choice for uncomplicated malaria as against chloroquine and sulfadoxine-pyrimethamine previously and; thirdly there was an increase in the production, procurement and distribution of long-lasting insecticide treated nets (ITNs) among homes in the sub-Saharan Africa (SSA).³¹

Due to these changes, the investigators sought to objectively determine the prevalence of congenital malaria in an urban and a semi-rural area in Lagos in 2014 because a mere inference from the above epidemiological changes may not be accurate. Placental histology was used in this study because some studies have shown that placenta histological examination is a more sensitive indicator for placenta infection than placental blood microscopy.³³⁻³⁴

METHODOLOGY

AIMS AND OBJECTIVES

General Aim:

The main aim is to determine the prevalence of congenital malaria among newborns delivered in an urban and semi-rural area in Lagos.

Specific objectives:

The specific objectives of this study are to:

1. Compare the prevalence of congenital malaria in preterm and term babies.
2. Compare the prevalence of congenital malaria between an urban area and a semi-rural area in Lagos.
3. Identify the risk factors for congenital malaria in neonates.
4. Correlate malaria parasite infection in maternal peripheral blood with the corresponding placenta tissue, cord blood and neonatal peripheral blood.
5. Compare the clinical and parasitological outcomes in the newborns of maternal use of Intermittent Preventive Treatment with Sulfadoxine-Pyrimethamine (IPT-SP) with non-use of IPT-SP.

SUBJECTS (MATERIALS) AND METHODS

This is a cross-sectional descriptive study carried out over a period of 7 months between April and October 2014 in two centres, Lagos university Teaching Hospital (LUTH), Idi-Araba and General Hospital (GH) Ijede, Lagos state. While Lagos state is a socio-cultural and commercial melting point attracting individuals from most tribes of the country. The population of Lagos is estimated to be at 16,536,000 in 2004³⁵. It has a tropical wet and dry climate and experiences two rainy seasons, with the heaviest rain

falling between April and July (about 400 mm per month on average) and weaker rainy season in October and November. The highest temperature is about 37.3⁰C. Malaria occurs all year round in the city and it is holo-endemic.

LUTH represents the urban arm of this study, and it is located in the south-western part of the country. It is a federal government-funded tertiary hospital in Lagos state located in Idi-Araba. It is 760- bedded and provides basic and specialist antenatal and obstetric care among other specialist services to the local community, patronized by people belonging to the various socioeconomic classes. Idi-Araba is one of the densely populated urban settlements in the state with poor sanitation in its surroundings. The neonatal unit of LUTH consists of an in-born section and an out-born section. The in-born section provides care for neonates delivered at the hospital while the out-born section provides similar care for babies delivered outside LUTH or who have been re-admitted after an initial discharge from the hospital. The neonatal unit is manned by four consultant paediatricians, 7–10 resident doctors and 4–6 medical interns. About 70 newborns are managed in the In-born section every month. About 196 neonates are delivered each month in LUTH on average.

General hospital (GH) Ijede on the other hand represents the semi-rural arm of this study and it is located 60km north-east of the state, overlooking the Lagos Lagoon with a population of about 8,208 people.³⁶ To the west, it is bordered by streams and swamp lands. The residents of Ijede are traditional fishermen who have been drawn into modern vocations and the urban life of Lagos. Ijede was selected because of its coastal location and because it has a health facility that enables the nature of subject enrollment required for this study in terms of obstetric and neonatal care compared to many other rural areas in Lagos. It is also a site for previous and ongoing malaria research General Hospital, Ijede is a 35-bedded secondary-level hospital and offers both antenatal and delivery services to the inhabitants among other services. The maternity ward has 10 beds and on average records 45 deliveries per month. The Obstetrics and Gynaecology department has a total of five doctors (one chief medical officer, one principal medical officer, two senior medical officers and one medical Officer) and four nurses. The paediatric ward for the older children has 10 beds with about 14 nurses and one medical officer who also takes care of the sick neonates in the maternity wing.

Inclusion criteria-

- Newborns delivered in LUTH and GH Ijede (regardless of their gestational age) and their respective mothers.

Exclusion criteria-

- Mothers who had not been living in Nigeria for at least 2 years prior to the study and their newborns.
- Mothers who withheld or could not provide consent and their newborns.

A total of 291 consecutive mothers who provided written informed consent with their corresponding babies were enrolled for this study. These were mothers who had been admitted either into the labour

ward or the ante-natal ward before delivery. A written informed consent was obtained from the mothers; those who delivered by planned Caesarian section and those who were not in labour at the time of enrollment gave a written informed consent before delivery while women who were already in labour signed the consent leaflet post-delivery after they had given a verbal informed consent before delivery (see appendix V). About 2 ml of venous blood was collected in Ethylene Diamine Tetra Acetic acid (EDTA) bottles from the mothers just before or within one hour after delivery. Same volume of blood was obtained from the cord blood at delivery while heel prick blood from the corresponding newborns were either directly smeared onto glass slides for thick and thin film preparation using standard techniques,³⁷ or 5–8 drops (about 0.5ml) of whole blood from heel prick of the newborns were collected into EDTA bottles, refrigerated, and later used to make thick and thick films. Blood samples collected that could not be processed immediately were kept in the refrigerator at temperature between 4-8⁰C in the neonatal unit at LUTH and the main laboratory of the GH Ijede. This set of samples were analyzed within 48hrs after collection.

Placenta tissues were also obtained from each mother. Biopsies (2cm by 2cm by 1cm) from the maternal side of the placenta were collected using a surgical blade and fixed in 10% neutral (phosphate) buffered formalin. Blood and placenta samples from a mother-newborn pair were labeled with similar codes to allow for identification. For instance, codes such as 4M, 4C, 4B and 4P referred to blood sample from the 4th mother, cord blood from the 4th newborn, heel prick blood from the 4th newborn and placenta tissue from the 4th mother respectively. The fixed placental samples were eventually transferred to the Anatomy and Molecular Pathology Department (Laboratory) of the Lagos University Teaching hospital, Idi-Araba, Lagos for processing.

After delivery and when the mother was deemed clinically stable, information on socio-demographics, antenatal history including use of intermittent preventative therapy, use of insecticide-treated net, recent use of antimalarial, peri-partum pyrexia and HIV status were obtained. Apart from direct questioning, the HIV status was further confirmed by checking through the mothers' hospital record since HIV screening is routinely done in both centres for all pregnant mothers who deliver in the hospitals. Only mothers who took IPT-SP under direct observation at the clinic were counted to have received IPT-SP. This information was captured using the questionnaire and the information obtained was kept confidential. The newborns were examined within one hour of delivery and the following anthropometric parameters were obtained: birth weight, length, head circumference (occipito-frontal circumference). These measurements were taken by the investigator and the research assistants whenever the investigator was not available. The individual measurements have been pre-validated before the study and the level of agreement was over 95%. The birth weights of the neonates were measured in grams using an electronic weighing scale (ADE, model: M11260); the babies' lengths were measured in centimetres using an Infantometer (Infalength) and the head circumferences in centimetres with an inelastic tape measure. The newborns were then classified based on their length and occipito-frontal circumference (OFC) values using the "Intra uterine growth standards for African infants" chart by Olowe.³⁸ The Ballard score chart was used to confirm the gestational age of the preterm babies. Term babies were defined as babies delivered at ≥

37 completed weeks while preterm babies were defined as those delivered before 37 completed weeks of gestation. Term babies were classified using their birth weights; SGA (< 2500g), AGA (2500-3999g) and LGA ($\geq$ 4000g). The Olowe's chart was used to classify the preterm babies into Appropriate for Gestational age (AGA), Large for Gestational Age (LGA) and Small for Gestational Age (SGA).³⁸

Preparation of thin and thick films and microscopic examination of the films for malaria parasites were completed at the Lagos University prepared at the WHO malaria research laboratory (ANDI centre of excellence) at the College of Medicine, University of Lagos, LUTH, Idi-Araba. At the laboratory both films were stained with 10% freshly prepared Giemsa stain maintained at a pH of 7.2. The stained blood smears were viewed under light microscope at x100 oil immersion magnification within 24-48hrs of preparation. The diagnosis of malaria was based on the identification of asexual stages of plasmodium on the thick blood smears, while thin blood smears were used to identify the species of plasmodium.

The microscopic evaluation of the slides (peripheral blood film) was carried out by two independent WHO certified level 1 microscopist for quality control as this is the standard operating procedure at the laboratory. The level of agreement in terms of parasite detection was 100% between the microscopists. Plasmodium parasite density (number of parasite per microliter of blood) was determined by counting the number of asexual parasites against 200 leucocytes on the thick blood film and converted to parasites per microliter of blood using an assumed total white blood count (WBC) of 8000cells/microliter.³⁹ For the purpose of this study, congenital malaria was defined as the identification of asexual forms of malaria parasite in the peripheral blood film of a new-born at birth. Cord blood and maternal blood microscopy was completed to correlate and confirm a trans-placental transfer of malaria infection from the mother to the newborn. Blood films were declared negative if no parasite was seen after viewing 200 high power fields. The recruited newborns were examined for clinical signs and symptoms of malaria throughout the period of admission.

The placental tissue processing was done at the Anatomic and Molecular Pathology Department of the Lagos University Teaching hospital, Idi-Araba, Lagos by the laboratory scientists and the histological slides were examined by a consultant pathologist. The placental tissues were processed using an automatic tissue processor. The tissues were then embedded in paraffin wax using an automatic tissue embedder, following which they were then cut into sections with the aid of a Microtome. The cut sections of each placenta tissue were then placed on a hot plate for about 30mins for the sections to adhere to the slides. Two different slides were made of each placental tissue: one for Haematoxylin &Eosin (H&E) staining and the other for Giemsa staining. The H&E slides were used to demonstrate the general structure of the placental tissue while the Giemsa-stained slides were used to identify malaria pigments.

A novel histological grading scheme for placental malaria by Muehlenbachs *et al*,⁴⁰ was adopted for this study. This grading system is a 2-parameter semi-quantitative grading scheme that scores degrees of inflammation and pigment deposition in placenta malaria. A diagnosis of placental malaria was made

only if malaria pigment with or without evidence of inflammation was present in a placenta tissue on histology.

DATA ANALYSIS

Categorical variables such as socio-economic status, educational level and parity were summarized as proportions while continuous variables such as gestational age were represented using mean and standard deviation. Univariate analysis involving categorical data was done using Chi Square test while difference between the means of two samples were tested using Student’s t test. A p-value < 0.05 was considered statistically significant. Statistical analyses were performed using Statistical Package for the Social Sciences Software (version 15; SPSS Inc., Chicago, IL).

RESULTS

A total of 291 mother-newborns pairs; 173 pairs (59.5%) from LUTH between April and October 2014, and 118 pairs (40.5%) from GH Ijede from April to July 2014 were enrolled for this study. The Lagos University Teaching Hospital, Idi-Araba, Lagos, represents the urban arm while General Hospital (GH) Ijede, Lagos, represents the semi-rural arm of the study. The majority (65.6%) of mothers were aged 25 to 34 years. More mothers from LUTH had tertiary education compared with GH Ijede (66.4% v 55.1%). Similarly, more mothers in LUTH belonged to socioeconomic classes 1,2 &3 compared to mothers in Ijede (82.2% v 68.6%) as shown in Table I according to Oyedeji’s socio-economic classification.⁴¹

The antenatal characteristics of the mothers are as shown in table II and Fig. 1 below. The parity of the mothers was similar across both centres and was almost equally divided among primiparous, secundiparous and para 3 (p = 0.221). HIV seropositivity rate was 5.2% in LUTH versus 1.7% in GH Ijede (p = 0.137). Fifteen of the total number of women enrolled (5.2%) had history of fever in the last two weeks of pregnancy and the proportion in both groups were similar (p = 0.965). A similar number (though not the same set of mothers) had treatment for malaria in the last two weeks of pregnancy (p = 0.965). As shown in Fig. 6, IPT-SP is the most used chemoprophylaxis during pregnancy in both centres with the percentage of the mothers from Idi-Araba being slightly higher than that of Ijede (87.8% vs 73.7%). None of the mothers from Idi-Araba used Chloroquine while a few proportions of the mothers from Ijede (4.2%) used chloroquine as chemoprophylaxis. The percentage of mothers who did not use any form of chemoprophylaxis is higher in Ijede compared to Idi-Araba (19.5% vs 9.3%).

TABLE I: Socio-demographic characteristics of mothers studied

Parameters	Total (%) n = 291 (%)	Idi-Araba n = 173 (%)	Ijede n = 118 (%)	P value*
Age(years)				0.0001
≤24	31 (10.7)	8 (4.6)	23 (19.5)	
25–34	191 (65.6)	117 (67.6)	74 (62.7)	
≥35	69 (23.7)	48 (27.8)	21 (17.8)	
Educational level				< 0.0001
Primary/none	10 (3.4)	6 (3.5)	4 (3.4)	
Secondary	101 (34.7)	52 (30.1)	49 (41.5)	
NCE/OND/HND	107 (36.8)	44 (25.4)	63 (53.4)	
University	73 (25.1)	71 (41.0)	2 (1.7)	
Social class				< 0.0001
Class I	42 (14.4)	38 (22.0)	4 (3.4)	
Class II	58 (20.0)	43 (24.8)	15 (12.7)	
Class III	123 (42.3)	61 (35.3)	62 (52.5)	
Class IV	67 (23.0)	31 (17.9)	36 (30.5)	
Class V	1 (0.3)	0 (0)	1 (0.9)	

OND = Ordinary National Diploma; HND = Higher National Diploma; NCE = National Certificate of Education. *P values for comparison between the various subgroups between Idi-Araba and Ijede.

TABLE II: Antenatal characteristics of mothers

Parameters	Total n = 291 (100%)	Idi-Araba n = 173 (59.5%)	Ijede n = 118 (40.5%)	P value*
Parity				0.221
Primiparous	86 (29.5)	46 (26.6)	40 (33.9)	
Secundiparous	82 (28.2)	46 (26.6)	36 (30.5)	
Para three	57 (19.6)	35 (20.8)	21 (17.8)	
Para four	46 (15.8)	30 (16.8)	17 (14.4)	
Grand multiparous	20 (6.9)	16 (9.2)	4 (3.4)	
Peripartum pyrexia				0.965
Present	15 (5.2)	9 (5.2)	6 (5.1)	
Absent	276 (94.8)	164 (94.8)	112 (94.9)	
HIV status				0.137
Positive	11 (3.8)	9 (5.2)	2 (1.7)	
Negative	280 (96.2)	164 (94.8)	116 (98.3)	
Malaria treatment ^{&}				0.965
Yes	15 (5.2)	9 (5.2)	6 (5.1)	
No	276 (94.8)	164 (94.8)	112 (94.9)	

[&]Malaria treatment in the last two weeks of pregnancy; *P values for comparisons between the various subgroups between Idi-Araba and Ijede.

With respect to other methods of malaria prevention, the use of indoor insecticide sprays and insecticide-treated nets were the most common methods used other than IPT-SP (Fig. 2).

Of the 39 women who did not use any chemoprophylaxis, nine of them used both ITN and indoor insecticide spray, 28 of them used either ITN or indoor insecticide sprays while only the remaining two mothers did not use any form of prophylaxis.

Characteristics of the newborns included in the study

Table III shows the characteristics of the neonates at birth. Thirty-two (11.0%) of the 291 babies were born preterm, with more preterms from LUTH than GH Ijede (16.2% v 3.4%, $p = 0.0006$). The mean ($\pm$ SD) birth weight of babies was 3059 ($\pm$ 627) g, and the mean gestational age was 38.3 ($\pm$ 1.9) weeks. The mean birth length was 48.0 ($\pm$ 3.4) cm while the mean head circumference was 34.9 ($\pm$ 1.8) cm. Majority

(78.1%) of the preterms were appropriate for gestational age. There were more SGA term newborn babies in Ijede and more term LGA newborn in LUTH. The mean head circumference was significantly smaller in newborns delivered in Ijede while the mean birth lengths were similar in both centres. One-hundred and forty-seven of the babies were males while the remaining 144 were females with a Male to Female ratio of approximately 1:1.

Malaria parasitaemia

Figure 3 shows the percentage of malaria parasitaemia in the various samples tested. Only one of the 291 mothers tested positive to malaria parasite by microscopy (0.34%) and the only positive result was obtained from the semi-rural area. The specie identified was *P. falciparum* with a parasite count of 25,786/ul. This mother was also reported to have used IPT-SP during pregnancy. All the babies and their corresponding cord blood samples were negative for malaria parasites by microscopy. However, 55 of the 291 (18.9%) placentae examined by histology demonstrated both malaria pigment and evidence of inflammation while two placenta tissues (0.7%) demonstrated evidence of inflammation only. The two women whose placenta tissue had only evidence of inflammation without malaria pigment deposition were from the urban area. Figure 4 shows the percentage parasitaemia in each of the centres. Maternal parasitaemia in Ijede was 0.85% while the placental parasitaemia rate was 43.2%. On the other hand, zero percent maternal parasitaemia and 2.3% placental parasitaemia was reported in Idi-Araba. Both centres recorded zero percent cord and neonatal parasitaemia.

Fifty-one of these infected placentae (92.7%) were from GH Ijede while the remaining four were from LUTH as shown in Table IV below.

TABLE III: Characteristics of the studied neonates.

Parameter	Total n = 291 (100%)	Idi-Araba n = 173 (%)	Ijede n = 118 (%)	P value*
Gender				0.271
Male	147 (50.5)	93 (53.8)	54 (45.8)	
Female	144 (49.5)	80 (46.2)	64 (54.2)	
Gestational age, mean (SD), weeks	38.3 (1.9)	38.1 (2.2)	38.6 (1.3)	0.023
Gestational age category				0.0006
Preterm	32 (11.0)	28 (16.2)	4 (3.4)	
Term	259 (89.0)	145 (83.8)	114 (96.6)	
Anthropometry at birth				
Birth weight, mean (SD), g	3059.3 (627.4)	3100.9 (678.9)	2998.3 (540.0)	0.171
Length at birth, mean (SD), cm	48.0 (3.3)	48.0 (3.6)	48.0 (2.7)	0.965
OFC at birth (SD), cm	34.9 (1.8)	35.1 (1.6)	34.7 (2.0)	0.041
Preterm	32	28	4	0.737
Small gestational age	4 (12.5)	4 (14.3)	0 (0)	
Appropriate for gestational age	25 (78.1)	21 (75.0)	4 (100.0)	
Large for gestational age	3 (9.4)	3 (10.7)	0 (0)	
Term	259	145	114	< 0.0001
SGA (< 2500g)	20 (7.8)	3 (2.1)	17 (14.9)	
AGA (2500-3999g)	222 (85.7)	130 (89.6)	92 (80.7)	
LGA (≥ 4000g)	17 (6.6)	12 (8.3)	5 (4.4)	
Length				0.911
< 2 SD	27 (9.3)	15 (8.7)	12 (10.2)	
Normal	249 (85.6)	149 (86.1)	100 (84.7)	
> 2 SD	15 (5.1)	9 (5.2)	6 (5.1)	
OFC	21 (7.2)	0 (0)	21 (17.8)	< 0.0001
< 2 SD	265 (91.1)	171 (98.8)	94 (79.7)	
Normal	5 (1.7)	2 (1.2)	3 (2.5)	

Parameter	Total n = 291 (100%)	Idi-Araba n = 173 (%)	Ijede n = 118 (%)	P value*
> 2 SD				

SD = Standard Deviation; OFC = Occipito-Frontal Circumference; * p values for comparisons between the various subgroups between Idi-Araba and Ijede.

Table IV: Placental malaria infection score and distribution

Parameter	Idi-Araba n = 173 (%)	Ijede n = 118 (%)	Total n = 291 (%)
Pigment score			
I	4 (2.3)	40 (33.9)	44 (15.1)
II	0 (0)	11 (9.3)	11 (3.8)
III	0 (0)	0 (0)	0 (0)
Total	4 (2.3)	51 (43.2)	55 (18.9)
Inflammation score			
I	7 (4.0)	40 (33.9)	47 (16.2)
II	0 (0)	9 (7.6)	9 (3.1)
III	0 (0)	1 (0.9)	1 (0.3)
Total	7 (4.0)	50 (42.4)	57 (19.6)

Factors associated with Placental Malaria

As shown in Table V, more of the mothers from the semi-rural area had placental malaria compared with mothers who delivered at the urban area (92.7% v 7.3%, $p = < 0.0001$).

About one-third of mothers aged ≤ 24 years enrolled in this study (11 out of 31) had placental malaria whereas only about one out of every six of the 69 mothers in the older age group ≥ 35 years enrolled in this study had placental malaria ($p = 0.044$). Only three of the seventy-three mothers with a university or higher level of education had placental malaria while four out of the ten (40%) mothers with no education or only primary level of education had placental malaria ($p = 0.001$).

The proportion of mothers belonging to the lower socioeconomic class were significantly represented in the group of mothers with placental malaria (23.4% v 34.5%) whereas those in the upper socio-economic class were less represented in the placental malaria group (76.6% v 65.5%, $p = \text{value } 0.030$). Furthermore,

the use of IPT-SP was a strong protective factor against placental malaria with only about 16% of mothers who used IP-SP having placental malaria whereas as high as 30% of the mothers who did not use IPT-SP had placenta malaria ($p = 0.016$). Though primiparous women had the highest prevalence of placental malaria (40%) this was not significantly different compared with mothers of other parities.

Table V: Risk factors for placental malaria

Characteristics	Total	Placental malaria		P-value*
	n = 291 (%)	Positive, n = 55(%)	Negative, n = 236 (%)	
Delivery site				< 0.0001
Idi-Araba	173 (59.5)	4(7.3)	169(71.6)	
Ijede	118 (40.5)	51 (92.7)	67 (28.4)	
Maternal age				0.044
≤24 years	31 (10.7)	11 (20.0)	20 (8.5)	
25–34 years	191 (65.6)	32 (58.2)	159 (67.3)	
≥35 years	69 (23.7)	12 (21.8)	57 (24.2)	
Parity				0.276
Primiparous	86 (29.5)	22 (40.0)	64 (27.1)	
Para two	82 (28.2)	12 (21.8)	70 (29.7)	
Para three	57 (19.6)	9 (16.4)	48 (20.3)	
Para four	46 (15.8)	10 (18.2)	36 (15.3)	
≥ Five	20 (6.9)	2 (3.6)	18 (7.6)	
IPT-SP status				0.016
Use	239 (82.1)	39 (70.9)	200 (84.7)	
Non-use	52 (17.9)	16 (29.1)	36 (15.3)	
Peripartum pyrexia				0.572
Present	15 (5.2)	2 (3.6)	13 (5.5)	
Absent	276 (94.8)	53 (96.4)	223 (94.5)	
HIV status				0.951
Negative	280 (96.2)	53 (96.4)	227 (96.2)	
Positive	11(3.8)	2 (3.6)	9 (3.8)	
Educational level				0.001
≥University	73 (25.1)	3 (5.4)	70 (29.7)	
NCE/OND/HND	107 (36.8)	26 (47.3)	81 (34.3)	
Secondary	101 (34.7)	22 (40.0)	79 (33.5)	
None/primary	10 (3.4)	4 (7.3)	6 (2.5)	

Characteristics	Total	Placental malaria		P-value*
	n = 291 (%)	Positive, n = 55(%)	Negative, n = 236 (%)	
SEC				0.030
Upper (I, II & III)	223 (76.6)	36 (65.5)	187 (79.2)	
Lower (IV & V)	68 (23.4)	19 (34.5)	49 (20.8)	

SEC = Socioeconomic class; * p values for comparisons between the various subgroups between Idi-Araba and Ijede.

There was no significant difference in the frequency of placental malaria among mothers with HIV and those without HIV infection (p = 0.951).

As shown in Fig. 5 below, two-hundred and thirty-nine (82.1%) of the mothers enrolled in this study used IPT-SP but only 39 (16.3%) of them had placental malaria compared to the five mothers who used chloroquine in which 4 (80%) of them had placental malaria. Twelve of the forty-two mothers (28.5%) who did not use any chemoprophylaxis or used local remedies had placental malaria and this represents 4.1% out of the 18.9% placental malaria prevalence observed in this study. Only five mothers used daraprim and none of them had placental malaria.

Table VI below shows a significant inverse relationship between the number of IPT-SP doses and the frequency of placenta malaria (p = 0.0048).

Table VI: Association between number of doses of IPT-SP and placental malaria.

Number of doses	Total	Placental malaria		P- value
	n = 291 (%)	Positive, n = 55 (%)	Negative, n = 236 (%)	
None	47 (16.2)	14 (25.5)	33 (14.0)	0.0048
One	66 (22.7)	19 (34.5)	47 (19.9)	
Two	177 (60.8)	22 (40.0)	155 (65.7)	
Three	1 (0.3)	0 (0)	1 (0.4)	

With similar gestational age, the babies of mothers with placental malaria had lower birth weight (2881.8 v 3100.7 g, p = 0.020) and head circumference (34.3 v 35.1 cm, p = 0.006) though the mean birth lengths were similar (Table VII).

Table VII: Association between placental malaria and neonatal outcome at birth

Neonatal parameters	Placental malaria		P-value*
	Positive	Negative	
Mean Gestational age (SD), weeks	38.3 (2.00)	38.3 (1.9)	0.837
Mean birth weight (SD), g	2881.8 (558.2)	3100.7 (636.4)	0.020
Mean birth length (SD), cm	47.6 (3.4)	48.1 (3.2)	0.306
Mean Head circumference (SD),cm	34.3 (1.6)	35.1 (1.8)	0.006

* p values for comparisons of the various subgroups between the group with placental malaria and those without placental malaria.

The associations between the use or non-use of IPT-SP with the outcomes in the newborn such as the gestational age, the birth weight, the length and the head circumference are as illustrated in table VIII below. There was no significant difference in the mean anthropometric measurements and the mean gestational age between mothers who used IPT-SP and those who did not use IPT-SP. The mean gestational age in the IPT-SP and no IPT-SP groups were 38.4 ($\pm$ 1.8) and 38.1 ($\pm$ 2.3) weeks respectively with a P value = 0.308. The mean birth weights were 3084 ($\pm$ 611.6) and 2941 ($\pm$ 689.3) gram respectively with a P value = 0.136. The mean birth lengths for both groups were also similar with values of 48.0 ($\pm$ 3.1) and 48.1 ($\pm$ 3.9) cm respectively. Also, the mean head circumferences were 35.0 ($\pm$ 1.7) and 34.6 ($\pm$ 2.1) cm respectively.

Table VIII: Association between maternal IPT-SP status and neonatal outcome

Parameters	Use of IPT-SP		P value
	Yes, n = 239 (%)	No, n = 52 (%)	
Mean Gestational age (SD), weeks	38.4 (1.8)	38.1 (2.3)	0.308
Mean Birth weight (SD), g	3084 (611.6)	2941 (689.3)	0.136
Mean Birth length (SD), cm	48.0 (3.1)	48.1 (3.9)	0.865
Mean Head circumference (SD), cm	35.0 (1.7)	34.6 (2.1)	0.106

DISCUSSION

The prevalence of congenital malaria in this study was zero. This finding is similar to the observed prevalence two decades earlier in a study at the urban arm of this current study by Lamikanra *et al*,²⁸ in 1993 where 2.97% of the mothers studied had malaria parasitemia but none of the newborns had malaria parasitemia. Similarly, Enweronu-Laryea *et al*,⁴² in 2010 in Ghana observed 0% congenital malaria prevalence using light microscopy among newborns delivered to 522 mothers. A similar result

was reported in Central India where Singh *et al*,⁴³ in a study carried out between 1992 and 1995, reported 0% prevalence of congenital malaria among newborns delivered to 2,127 women. A few other researchers have also reported very low prevalence of congenital malaria; In Sagamu, Southwest Nigeria, Sule-Odu *et al*,⁴⁴ reported a prevalence of 0.7% in 2002, while Mwaniki *et al*,⁴⁵ in an 8-year study in Kenya published in 2010 documented that only 0.35% of 4,790 newborns had congenital malaria.

On the other hand, the result from this study is in contrast with the prevalence rate of 15.3% as determined by Mukhtar *et al*,¹⁴ in Idi-Araba, the same urban area used in the present study, about a decade earlier. This significant change over the previous ten years may be due to several synergistic factors in the control of malaria. Between these two periods, there has been a rapid scale up in preventive measures such as the use of insecticide treated nets and the use of IPT-SP.³¹

As reported in the current study, over 40% and 80% of the mothers used ITN and IPT-SP respectively during pregnancy. The present study shows that the use of IPT-SP during pregnancy reduces the risk of malaria in pregnancy which subsequently lowers the risk of congenital malaria. In addition, the widespread use of the highly effective artemisinin-combination therapy for the treatment of acute cases of malaria in place of chloroquine may have contributed to the rapid decline in the prevalence of congenital malaria observed over this period.

Placental sequestration of malaria parasites as observed in this study may also be responsible for the absence of microscopically detectable malaria parasites in the neonatal peripheral blood film.¹⁰ Another possible reason for the low/absence of congenital malaria in the present study compared to earlier studies may be due to the characteristics of the study population. While the present study recruited consecutive mothers and their newborns, some of the previous studies with a high prevalence of congenital malaria have selected women with confirmed peripartum malaria infection while some others recruited unwell babies with signs and symptoms of infection, thus resulting in a significantly high prevalence of neonatal parasitaemia.^{23,26,29}

The prevalence of cord parasitaemia in this study was 0%, corresponding with the 0% prevalence of congenital malaria. This is also very different from the findings in some of the previous studies.^{21,28} However, this is not too surprising as the incidence of maternal parasitaemia in the current study was also very low.

The prevalence of maternal parasitaemia in this study was 0.34% (1 out of 291). This maternal parasitaemia prevalence is significantly low compared to most of the previous studies.^{22,28} However, some earlier studies have also likewise reported low prevalence rates of maternal malaria. An example is the study by Parekh *et al*,⁴⁶ who reported a maternal parasitaemia prevalence of 1%. Another example is the study by Ahmed *et al*,⁴⁷ published in 2014 in which the maternal malaria prevalence was 1.8% among 2,282 women at the time of delivery. As in the case of congenital malaria, this finding of low maternal parasitaemia most likely indicates the effectiveness of the current malaria preventive strategies. Although some of these previous studies were also undertaken during IPT-SP era, this current study is

more recent and IPT-SP among other strategies has gained better acceptance than before.³⁸ This low maternal parasitaemia could also be as a result of placenta sequestration of malaria parasites as mentioned earlier.^{10,48} However, some of these women (5.2%) admitted to taking one form of anti-malaria or the other in the later part of their pregnancy which may have also reduced the chances of a microscopically detectable malaria parasitaemia.

The rate of placental malaria parasitaemia (18.9%) observed in this study as evidenced by the demonstration of malaria pigments on placenta histology is close to that of 22% reported by Parekh *et al*,⁴⁶ in a study done in Peru in 2010. This finding indicates that these women have had a clinical or sub-clinical malaria infection during their pregnancies. This placental malaria infection rate did not correlate with any of the maternal, cord or neonatal parasitaemia. More so, the placenta tissue, cord blood and neonatal blood of the only positive maternal peripheral blood were all negative for malaria. This lack of correlation is similar to what was observed by parekh *et al*.⁴⁶ In that study, only two out of 193 (1%) women had malaria parasitaemia by microscopy whereas 22% of the placentae examined after delivery revealed malaria pigment on histology. The implication of this finding is that the measurement of acute infection at the time of delivery by blood sampling may not provide an accurate measure of malaria exposure or experience throughout the course of pregnancy. Some other studies have also demonstrated a similar finding of lack of correlation between maternal parasitaemia and placenta parasitaemia.^{15,23,40,49} This further strengthens the hypothesis that placental sequestration of malaria parasites does occur during pregnancy, leading to microscopically undetectable malaria parasites in the peripheral blood films. However, this finding is different from that observed by some other researchers where maternal parasitaemia correlates well with placental, cord and neonatal parasitaemia.²¹ This difference can be explained by differences in methodology as these latter studies where correlations were reported utilized placenta blood microscopy while this current study, just like the former ones, utilized placenta tissue histology.

The risk factors and the clinical features of congenital malaria could not be described in this study since there was no case of congenital malaria. However, since placental parasitaemia has been proven to be a risk factor for congenital malaria, the risk factors for placenta malaria have been described instead.^{22,50,51}

As expected, the prevalence of placental malaria was significantly higher in the semi-rural area of Ijede than in the urban area of Idi-Araba. This rural prevalence of placental malaria in this study is similar to what was observed by Beaudrap *et al*,⁵ in a study among some Ugandan women published in 2013. This may be related to the coastal location of Ijede which favours the breeding of the malaria parasite's vector. Moreover, being a semi-rural area, access to health facilities is relatively less compared to the urban arm of this study.

However, despite significantly higher placental malaria prevalence in the semi-rural area compared to the urban area, there was no difference in the prevalence of congenital malaria between the two areas. This could be explained by the sequestration of malaria parasites in the placenta, which suggests that the

placenta plays a significant role, serving as a barrier to the occurrence of congenital malaria in newborns. The use of IPT-SP was also significantly associated with a reduced prevalence of placental malaria; the mothers who had only one dose of IPT-SP or no dose were more likely to have placental infection. This association with IPT-SP use is similar to that observed by Anchang-Kimbi *et al*,³⁴, Kolawole *et al*,¹⁵, and Catherine *et al*,²³. There was also an association between the number of doses of IPT-SP and frequency of placental malaria as placental malaria was less prevalent in mothers who took at least two doses of IPT-SP when compared with those who took only one dose. This is similar to the findings by Anchang-Kimbi *et al*,³⁴. This present study supports the recommendation that at least two doses of IPT-SP should be taken in pregnancy in order to reduce the burden of malaria in pregnancy.

Another significant association occurred between maternal level of education and placental malaria. The prevalence of placental malaria is much less in the more educated category of mothers. This is identical to what was observed by Mukhtar *et al*,¹⁴. This is probably because the more educated mothers are better abreast of the malaria preventive measures, and they are more likely to seek medical attention compared to their less educated counterparts. Rural settlement is obviously a risk factor for placenta malaria as most of these infected placentae were derived from the mothers in the semirural area of Ijeda. The socio-economic status, which correlates with the maternal educational status, also significantly affected the prevalence of placental malaria. This observation has been made previously by Kolawole *et al*,⁵² in Ilorin, Nigeria. This may be related to the housing conditions and access to healthcare services.

Younger women were more likely to have placental malaria compared with older women. This finding most likely reflects the increased predisposition of primiparous women to malaria in pregnancy as younger women are more likely to have a lower parity and lower immune experience than older women. This finding may be related to a relative immune inexperience in the younger group of mothers. Although primiparous women were more likely to have placental malaria in the present study the difference was not statistically significant.

There was no significant association between placental malaria and peripartum pyrexia in this study. This is not in agreement with the finding by Sotimehin *et al*,⁵³ in a study on the possible risk factors for congenital malaria in a tertiary care hospital in Sagamu. However, Sotimehin *et al*,⁵³ considered the presence of fever in the last three months of pregnancy while this study considered fever only in the last two weeks of pregnancy. With the declining burden of malaria in the general population it is not unexpected that the symptom of fever may be more likely due to other non-malaria causes. Similarly, HIV infection was not associated with placental malaria. This observation is similar to that of Rogerson *et al*,³³ but different from the findings by Perrault *et al*,⁵⁴ and Newman *et al*.⁵⁵ This difference may be a function of the degree of immune suppression in the HIV positive women rather than a mere seropositivity for the HIV virus.

Placental malaria in this study did not show significant association with the babies' maturity. This agrees with previous studies that had suggested that prematurity is more associated with maternal peripheral

parasitaemia rather than placental parasitaemia.²⁴ It was hypothesized that premature birth resulting from malaria in pregnancy is majorly due to the systemic inflammation induced in the mother by the malaria parasite but in this study the prevalence of maternal parasitaemia was very low. Therefore, premature deliveries in this study are most likely due to some other factors rather than malaria in pregnancy.

The present study adds to the evidence that malaria in pregnancy is not a benign condition but has adverse effects on the neonates. In the present study mothers with placental malaria had newborns with lower birth weight and head circumference despite similar gestational age at birth. The inverse association between malaria in pregnancy and birth weight has been documented previously by Falade *et al*,²⁴ and Rogerson *et al*.³³ Malaria in pregnancy may affect fetal growth in various ways. Firstly, by way of causing premature birth, babies born preterm are more likely to have a lower birth weight than babies born at term. Secondly, due to parasite sequestration and inflammation in the placenta, placental function is impaired leading to fetal malnutrition. However, this is different from the findings by Lamikanra,²⁸ Mukhtar *et al*,¹⁴ Kolawole *et al*,⁵² Anchang-Kimbi *et al*,³⁴ and Alfredo *et al*,⁵⁶ where no significant difference was observed. These differing findings by different studies may be a function of the degree and intensity of placenta sequestration and inflammation. Leopardi *et al*,⁵⁷ in 1996 suggested that only active placenta malaria and not past infection had a negative effect on fetal growth. It is unclear what the implication of the reduced anthropometric measures at birth is on the long-term growth and neurological development of these children; however, there is a need for close monitoring of these newborns.

With respect to the outcome of the use and non-use of IPT-SP on the parasitological outcome of the babies, there is no difference in both groups as all the heel prick samples from the babies enrolled were negative. This finding may be due to placental sequestration of malaria parasite as evidenced by the finding of malaria pigment deposition in about one-fifth of the women enrolled in this study. Placental sequestration, as seen in this study, has been found to be responsible for microscopically undetectable malaria parasites in the peripheral blood film.¹⁰

The present study found no significant association between the newborn anthropometric measurements and the use of IPT-SP. This was different to the findings of Suleiman *et al*,⁵⁸ in a case control study involving 113 pregnant women in Mediani-Sudan. However, in that study only primigravidae were recruited, a subgroup known to be more at risk of malaria in pregnancy and which may benefit most from malaria preventive measures.

Overall, this study has demonstrated a significant decline in the prevalence of congenital malaria when compared with the results of many studies from various regions of Nigeria conducted in the previous one decade.^{14,15} In addition, this study identified residence in semirural area, younger maternal age, lower educational and socioeconomic status and non-receipt of IPT-SP as factors associated with placental malaria. The present study has also shown that malaria in pregnancy may be associated with adverse effect on fetal growth that may require follow-up.

CONCLUSIONS

1. There was a zero percent prevalence of congenital malaria in the present study. This shows that the prevalence of congenital malaria has significantly reduced compared to the previous decade.
2. There was a very low rate of maternal and neonatal parasitaemia observed in this study. This suggests that the recent scaling up of malaria prevention, treatment and control program using IPT-SP among other strategies has been effective in reducing the burden of malaria in pregnancy.
3. There was a comparatively higher prevalence of placental malaria among younger mothers, those with lower educational and socioeconomic status, mothers who received less than two doses of IPT-SP in pregnancy, and mothers residing in the semirural area.
4. This study shows that malaria in pregnancy, as reflected by a significant prevalence of placental malaria, has an adverse effect on the birth weight and the head circumference of the affected newborn.

LIMITATION(S) OF THE STUDY

1. Most of the women recruited in this study were “apparently healthy pregnant women” therefore reducing the chances of detecting malaria parasitaemia as only few of them had peripartum pyrexia.
2. The nutritional status of the enrolled mothers was not predetermined so the possible impact on the babies’ anthropometric parameters is not known.

Abbreviations

AGA Appropriate for Gestational Age

ACTs Artemisinin-based Combination Therapies

ANDI African Network of Drugs and Diagnostics Innovation

CSA Chondroitin Sulphate A

GA Gestational Age

IgG Gamma Immunoglobulin

IPT-SP Intermittent Preventive Treatment with Sulfadoxine-Pyrimethamine

ITN Insecticide Treated Nets

LBW Low Birth Weight

LGA Large for Gestational Age

MiP	Malaria in Pregnancy
OFC	Occipito-Frontal Circumference
P. falciparum	Plasmodium falciparum
PM	Placenta Malaria
RDTs	Rapid Diagnostic Tests
SGA	Small for Gestational Age
SP	Sulfadoxine-Pyrimethamine.
WHO	World Health Organization

Declarations

Ethics approval and consent to participate-Ethical approval for this study was received from the Ethical committee of the Lagos University Teaching Hospital (LUTH), Idi-Araba, Lagos state, Nigeria, and the Ethical committee of the General Hospital Ijede, Lagos state, Nigeria. Consents for this study were obtained from the mothers. We can **confirm that all experiments were performed in accordance with relevant guidelines and regulations.**

Consent for publication- No individual identifiable details are stated in this study.

Availability of data and materials- The datasets generated and/or analyzed during the current study are not yet publicly available because they have not been published yet but are available on request.

Competing interests- No competing interests.

Funding- This study was partly self-funded by the lead author with support from the WHO Malaria research Laboratory at the Lagos University Teaching Hospital, Idi-Ara, Lagos state, Nigeria.

Authors' contributions:

First Author - M.O is the lead author. He developed the hypothesis and obtained ethical approval from the two hospitals involved in this study. He arranged the collection and transport of the blood and placental samples from the study sites. to the respective laboratories for analysis. He also arranged the consents from the participants. He is the main writer of this article.

The second author-FEA-Is the second author and the lead supervisor. He supervised the activity of and provided guidance to the first author. He supervised the final write-up of the project.

Third Author- IC -He provided guidance for the author, assisted the lead author with guidance for the write up and interpretation of data.

Fourth Author-IB- He supervised the activity and provided guidance for the first author.

Fifth Author- W.O- Supervised the malaria microscopy and provided advice regarding the development of the thesis.

Sixth Author- A.D- Supervised the placental histological examination.

All authors have read and approved the final manuscript.

ACKNOWLEDGEMENTS

I wish to thank all my trainers who have impacted me one way or the other throughout my Residency training period. To Prof. Afolabi Lesi, thank you for being a great teacher and mentor. To Dr. Christopher Esezobor, I want to especially thank you for guiding me all the way in this study. To Prof. Wellington Oyibo, your wealth of experience and knowledge in malaria research and microscopy has really assisted in this project. I am glad you are a part of this study. To Dr. Ireti Fajolu and Prof. Edna Iroha, I am deeply grateful for guiding me into the field of malaria research and for your invaluable contributions throughout the study. I also want to appreciate the microscopists at the WHO Malaria Research Laboratory at LUTH for their contribution in terms of malaria microscopy.

Moreover, I appreciate Dr. Adetola Daramola for all her efforts in placenta histology. I thank the entire staff of the labor ward and neonatal ward for their co-operation throughout the study period. I want to particularly thank the residents, medical officers, house-officers, and nurses who rotated through the neonatal unit during my study in both centres for their assistance and support. Finally, but not least, I want to thank all the mothers and their corresponding babies who participated in this study. Without them this study would never have existed. Thank you for making yourselves available for this study.

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Figures

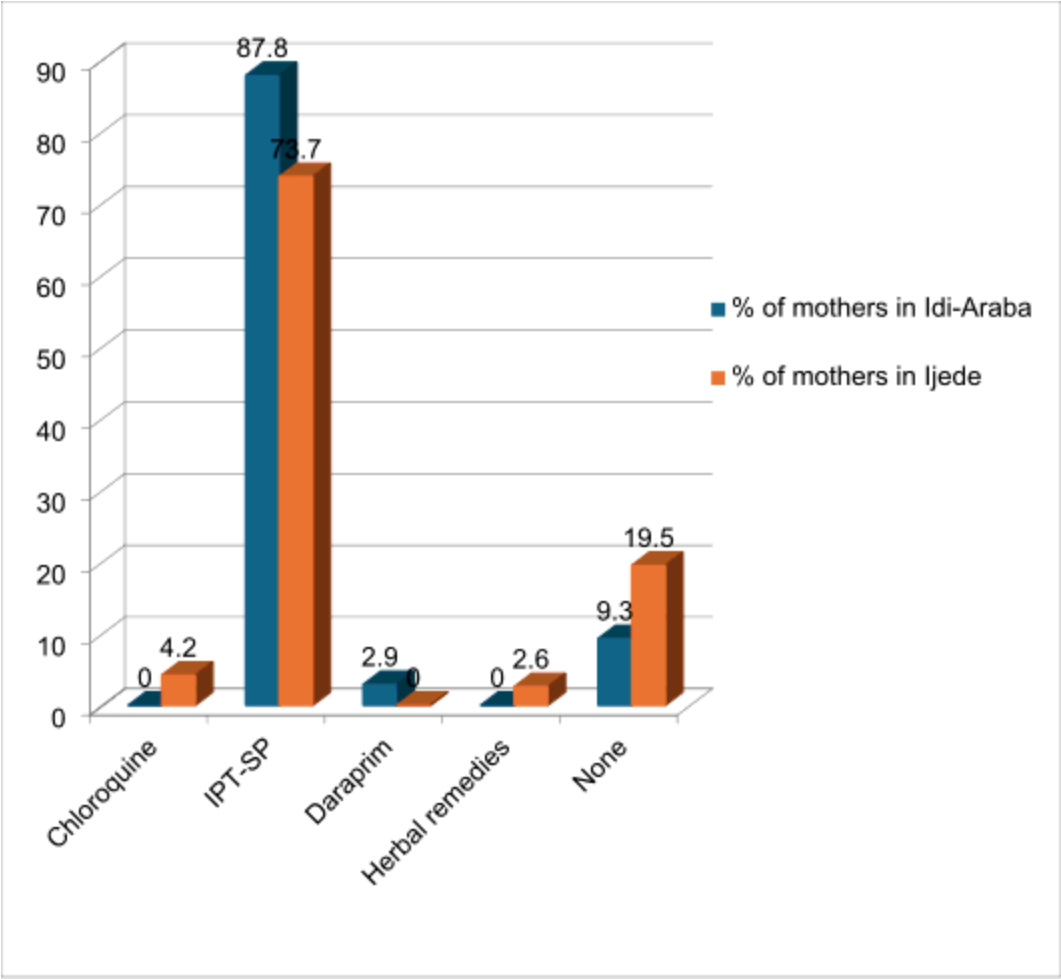


Figure 1
Distribution of Chemoprophylaxis methods across both centres.

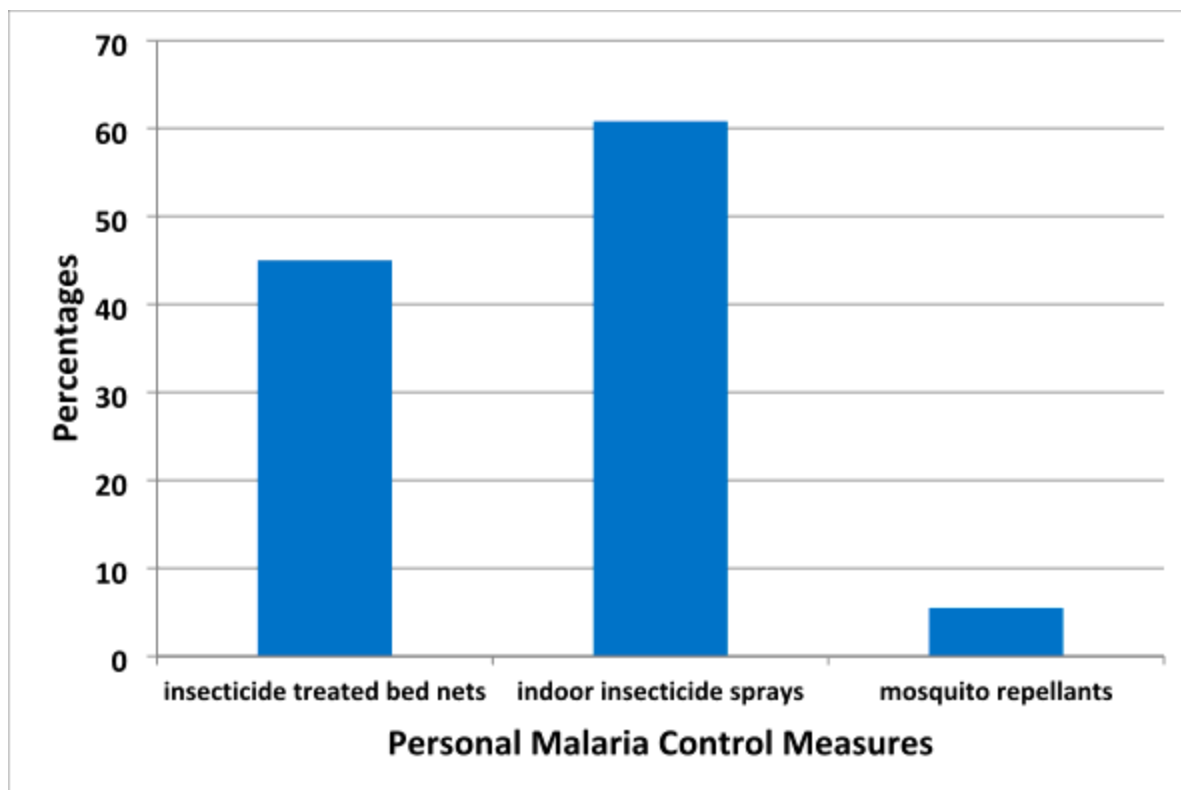


Figure 2

Distribution of the other methods of malaria prophylaxis in both centres.

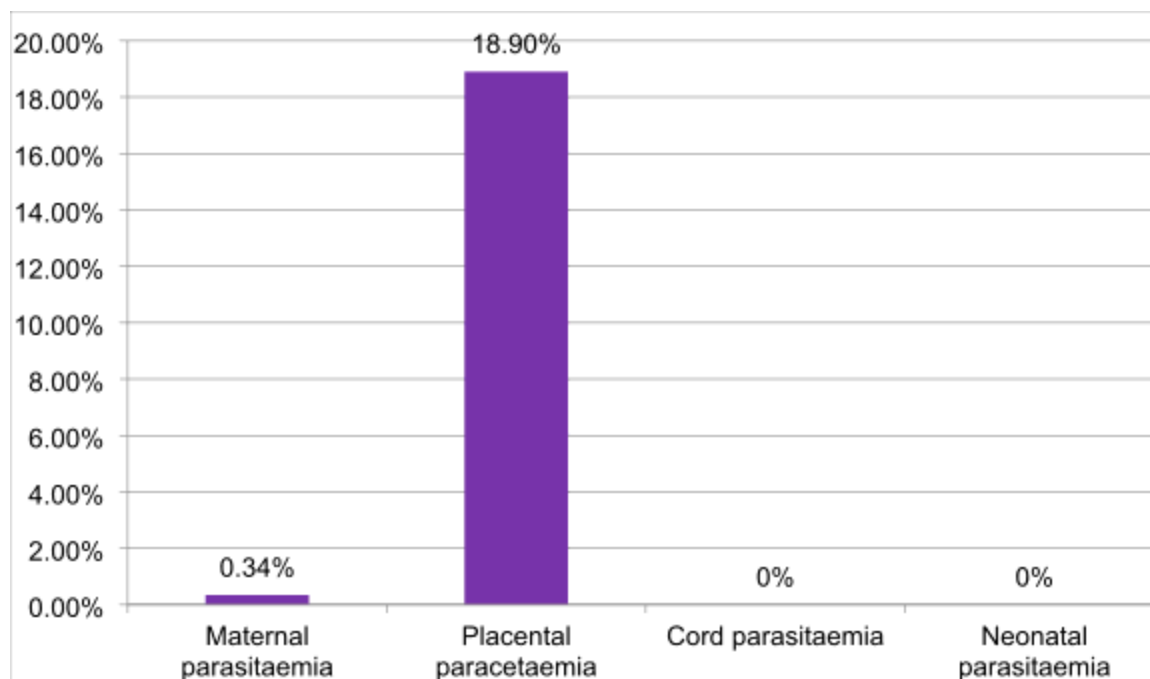


Figure 3

Malaria parasitaemia in both centres combined

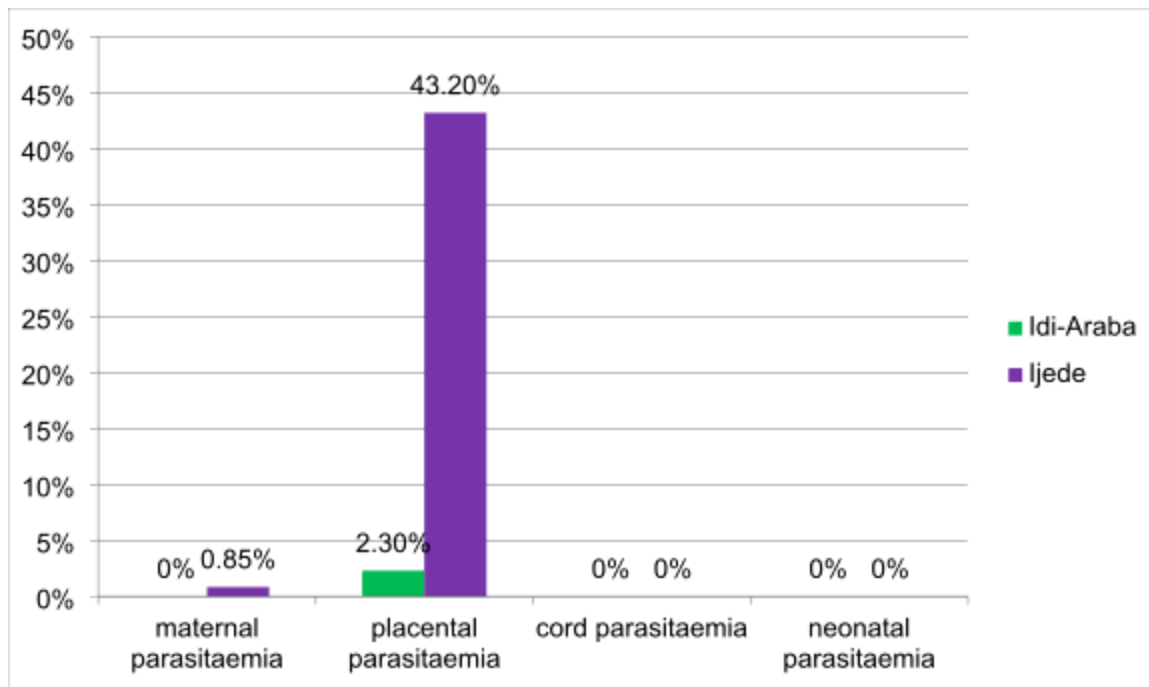


Figure 4

Malaria parasitaemia in each individual centre.

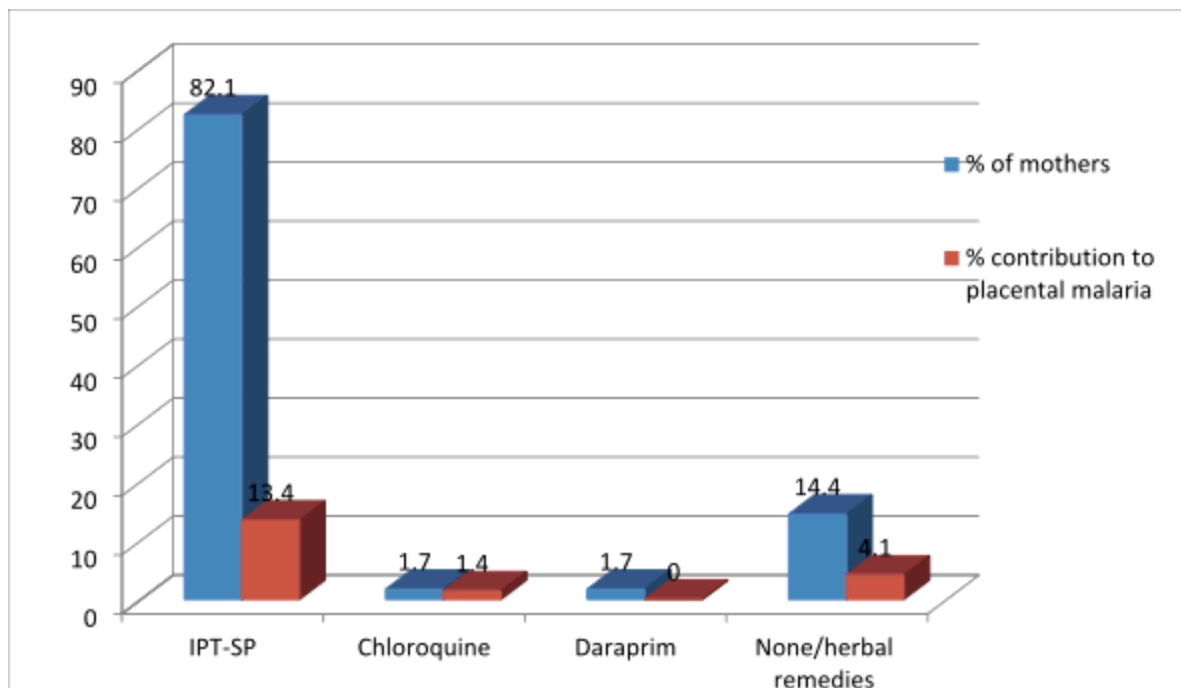


Figure 5

Prevalence of placental malaria in relation to methods of chemoprophylaxis.